Small-root reach with unequal bounds: no convex-monomial-set lattice beats 1/W, and the symmetric e-form asymptotes are optimal

2026-10-29

Small-root reach with unequal bounds

2026-10-29 · lead `LEAD-ICP-coppersmith-smallroot-semaev` · follows `research/smallroot_optimal_reach_20261026` (equal bounds, numerical search for unequal bounds) and §7c of `research/prime_s5_small_root_leads_20261028` (item 5 of the 2026-10-29 lead list: the unequal-bound e-form case left open there).

Class: analysis (a derived proof, exact integer counts and exact finite-field supports). Not an index-calculus measurement, not an improvement, no leaderboard row, no dashboard point. Track: algebraic (small-root decomposition); it ratchets nothing.

Result

The cited study proved a reach bound for **equal** per-variable bounds and could only search numerically when the bounds are unequal (best region 0.4357, rising towards 1/2, for S_3). This note closes that gap for the setting both studies work in.

- **[derived] Theorem K.** Let `f` have exponent set `F`, let `M` be the lattice points of a convex region in the orthant, and let `W_i = max_F ν_i − min_F ν_i` be the extent of `f` in coordinate `i`. Then the translate count that bounds every lattice gain satisfies `W_i · E(M) ≤ c_i(M) = Σ_{μ∈M} μ_i`. The proof is short and elementary (§2): a lattice translation into two fibres of `M`, a layer-cake sum, and a counting lemma.
- **[derived] Every bound direction at once.** With bounds `p^{δ_i}` the Howgrave–Graham condition `Σ_μ ⟨δ,μ⟩ < E(M)` then gives **`Σ_i δ_i W_i ≤ 1`**, so **`ε = Σ δ_i ≤ 1 / min_i W_i`**, for every direction of the bounds, every convex `M`, every level `t`. This is the bound the numerical search only approached: S_3 has `W_i = 2` and the supremum is 1/2.
- **[derived + measured supports] The symmetric e-form is closed at the Jochemsz–May asymptote.** In the e-variables `e₁..e_m` with weights `δ_{e_j} = j·δ_B`, the supports of S_3, S_4, S_5 are the **full simplices** with `W_j = D = 2^{m−1}` for every `j` (GP, §3). Hence **`δ_B ≤ 2 / (m(m+1)·2^{m−1})`: 1/6, 1/24 and 1/80 for `m = 2, 3, 4`**, exactly the asymptotes that §3.2 of the lead-4 study derived for one lattice family. No other lattice built from `(p, f)^t` on a convex monomial set does better.
- **Distance to the targets, unchanged and now unconditional for this class:** S_4 is **6×** short of δ = 1/4 and S_5 **16×** short of δ = 1/5 in the e-form, as in the table of §3.2 of the lead-4 study.

- **Not covered, and not claimed (§5):** multi-target (non-principal) ideals, and non-convex monomial sets. For each the exact step of the proof that fails is named.
- **Sharpness.** The constant 1 in Theorem K is sharp for the translate count: thin slabs reach 0.991 (Figure 1). Whether real lattices attain the translate count in that regime is **not checked** here (it is, exactly, in the 11 cases of the cited study; §4).
- **Lean:** the counting lemma, the direction-free reach inequality and the e-form instances are checked in core Lean (no `sorry`, standard axioms). Theorem K's fibre step, layer-cake step and the Newton-polytope step are **not formalized** (proof map, §6).

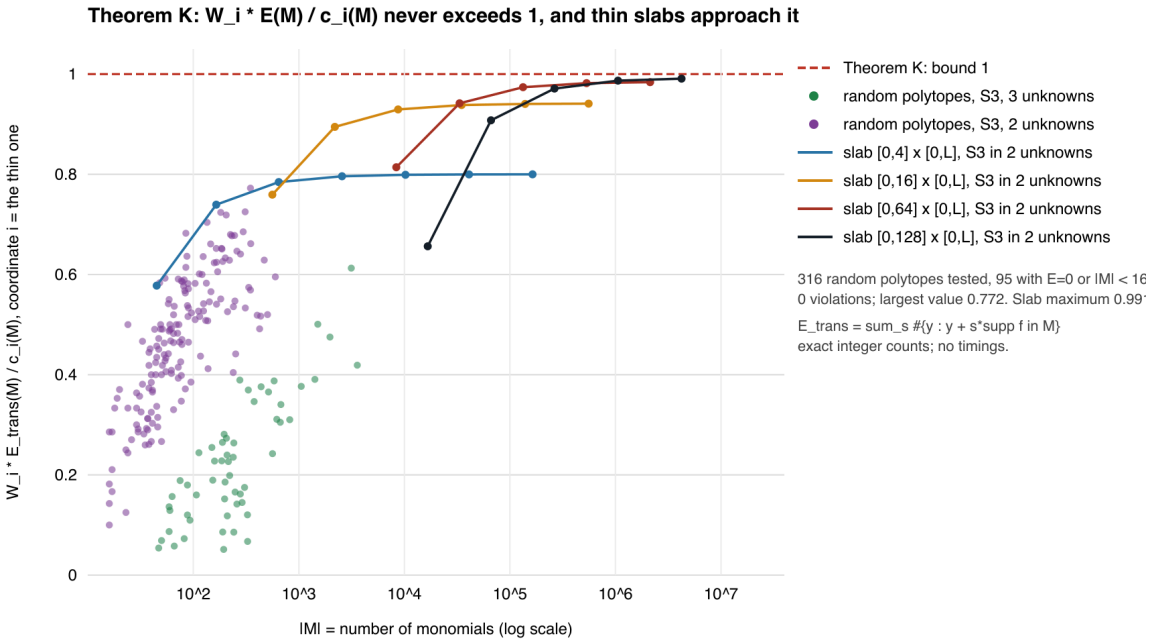


Figure 1: W_i·E_trans/c_i over random convex monomial sets and over thin slabs; Theorem K bounds it by 1. Source: reach_sharpness.svg.

0. Setting, and what is inherited

A block is `f ∈ F_p[x₁..x_n]` with bounds `|x_i| < p^{δ_i}`. The reach is `ε = Σ δ_i`, as in the cited study. For a monomial set `M`, `E(M) = log_p[L(M) : p^t Z^M]`, and Howgrave–Graham with dimension factors dropped (which favours the attacker) reads `Σ_{μ∈M} ⟨δ,μ⟩ < E(M)`.

Inherited from the cited study, not re-derived here `[derived there]`:

1. the layer bound `E(M) ≤ Σ_{s=1}^{t} dim(f̄^s F_p[x] ∩ span M)` (regular sequence `p, f`);
2. Ostrowski: `Newt(h·f^s) = Newt(h) + s·N`, `N = Newt(f)`. For `M = R ∩ Z^n` with `R` convex it follows that `supp h ⊆ {y : y + sN ⊆ R}`, so `dim(f̄^s F_p[x] ∩ span M) ≤ #{y ∈ Z^n_{≥0} : y + s·F ⊆ M}`.

Write $E_{tr}(M) = \Sigma_{s\geq 1} \#\{y \in Z^n_{\geq 0} : y + s\cdot\nu \in M \text{ for every } \nu \in F\}$ (the translate count). Then $E(M) \leq E_{tr}(M)$. Independent check of this layer of the argument: my own implementation of E_{tr} reproduces **all 11** exact lattice gains of the cited study (§4).

1. The direction reduces to one coordinate

[derived] Fix M, write $c = (c_1..c_n)$, $c_i = \Sigma_{\mu\in M} \mu_i$. The Howgrave–Graham condition is the *linear* constraint $\langle\delta,c\rangle < E$ on $\delta \geq 0$, and $\varepsilon = \Sigma\, \delta_i$ is linear. So the maximum of ε over all bound directions sits at a vertex: all the bound on one coordinate. That is what the cited study's optimiser kept finding (the whole bound on one variable, §3 there), now derived rather than observed. It also means the unequal-bound problem is exactly the family of single-coordinate ratios $E(M)/c_i(M)$.

2. Theorem K and its proof

Theorem K. [derived] Let $F \subset Z^n_{\geq 0}$ be the exponent set of f and $M = R \cap Z^n_{\geq 0}$ for a convex R. For a coordinate i let $m = \min_F \nu_i$, $M' = \max_F \nu_i$, $w = M' - m \geq 1$. Then

$$w \cdot E_{tr}(M) \leq \Sigma_{\mu\in M} \mu_i.$$

Proof. Let $\ell(u) = \#\{\mu \in M : \mu_i = u\}$ and $A_s = \{y \in Z^n_{\geq 0} : y + s\cdot\nu \in M \text{ for all } \nu \in F\}$.

1. *Fibre step.* Pick $\nu_{min}, \nu_{max} \in F$ attaining m and M'. The maps $y \mapsto y + s\cdot\nu_{min}$ and $y \mapsto y + s\cdot\nu_{max}$ are injective, send A_s into M, and shift the i-th coordinate by $s\cdot m$ and $s\cdot M'$. Hence for every t: $\#\{y \in A_s : y_i = t\} \leq \min(\ell(t+sm), \ell(t+sM'))$.
2. *Pairs.* Sum over t and put $u = t + sm$: $\#A_s \leq \Sigma_u \min(\ell(u), \ell(u + sw))$. Summing over $s \geq 1$ counts each pair $u < v$ with $v - u$ a positive multiple of w once: $E_{tr} \leq \Sigma_{u<v,\ w \mid v-u} \min(\ell(u), \ell(v))$.
3. *Layer-cake.* $\min(\ell(u), \ell(v)) = \Sigma_{\lambda\geq 0} 1[\ell(u)>\lambda]\cdot 1[\ell(v)>\lambda]$. With $S_\lambda = \{u : \ell(u) > \lambda\}$ the right side is $\Sigma_\lambda P_w(S_\lambda)$, $P_w(S) = \#\{(u,v) \in S^2,\ u < v,\ w \mid v-u\}$.
4. *Counting lemma.* For a finite $S \subset Z_{\geq 0}$: $w\cdot P_w(S) \leq \Sigma_{u\in S} u$. Split S into residue classes mod w; the j-th smallest element of a class is at least $j\cdot w$, so a class of size k has sum at least $w\cdot C(k,2)$, which is w times its pairs.
5. $\Sigma_\lambda \Sigma_{u\in S_\lambda} u = \Sigma_u u\cdot\ell(u) = c_i$. ∎

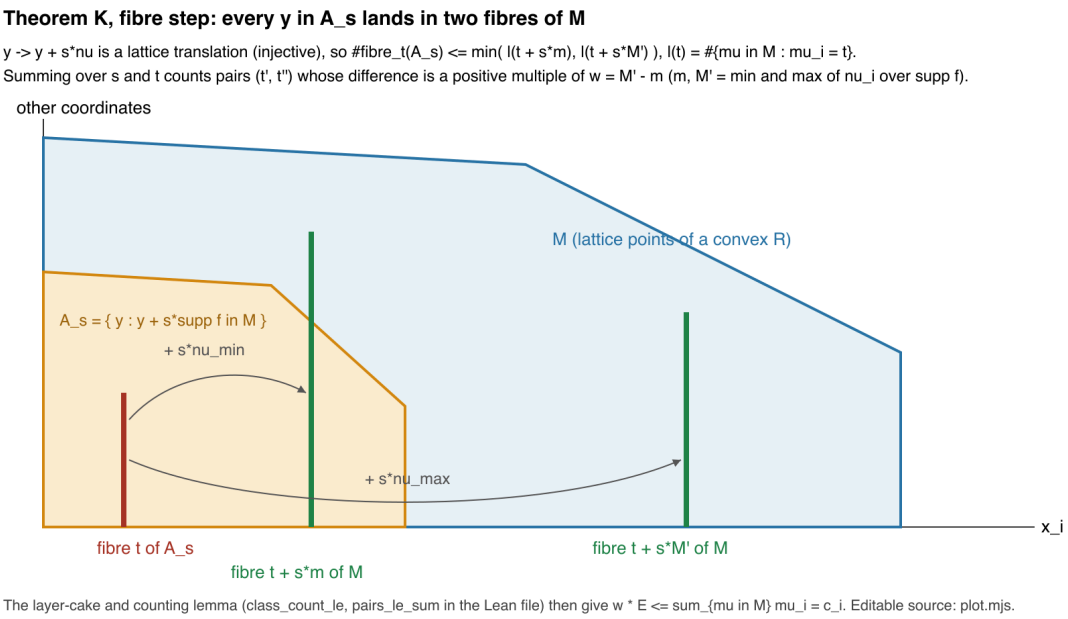


Figure 2: The fibre step: each y of A_s lands in the fibres t+sm and t+sM' of M. Editable source: plot.mjs; Lean: class_count_le, pairs_le_sum.

What the proof uses. Only that `M` contains `y + s·ν` for the two extreme exponents, which is where convexity enters (via Newton polytopes, §0). It does not use the dimension of `M`, the level `t`, or the shape of `F` beyond `m` and `M'`.

Corollary (every direction). `[derived]` If `E(M) > 0`, `δ ≥ 0` and `⟨δ,c⟩ ≤ E(M)`, then `Σ_i δ_i W_i · E ≤ ⟨δ,c⟩ ≤ E`, so **`Σ_i δ_i W_i ≤ 1`** and **`ε ≤ 1/min_i W_i`**.

Calibrations. It reproduces both known tight cases: a univariate polynomial of degree `d` has `W = d` and `δ ≤ 1/d` (Coppersmith); a linear form in two unknowns has `W₁ = W₂ = 1` and `δ₁ + δ₂ ≤ 1`. It is consistent with the bound of the cited study for equal bounds (which is smaller, because it uses the volume as well) and with the numerical optimiser (every value below 1/2). I did not search the literature for earlier statements of Theorem K; treat it as derived here, not as new.

3. The e-form and the targets

`[measured, exact]` Supports computed in PARI/GP 2.17.3 over `F_P`, `P = 2^31 − 1`, `a = −3`, `b` and `z` fixed random values: `S₃` by Semaev's formula, `S₄` and `S₅` by the resultant recursion, the e-form polynomial `T(e₁..e_m; z)` fitted at random points by solving a linear system with 8 spare equations as a check (`eform_support.gp`, output `eform_support.out`):

m	polynomial	D	monomials in e	nonzero	max exponent of e_j	min	e_j^D present
2	$S_3(e_1,e_2;z)$	2	6	6	2, 2	0, 0	yes, yes
3	$S_4(e_1,e_2,e_3;z)$	4	35	35	4, 4, 4	0, 0, 0	yes ×3
4	$S_5(e_1..e_4;z)$	8	495	495	8, 8, 8, 8	0 ×4	yes ×4

A nonzero coefficient at a random point is a certificate that the monomial is present for generic `(a, b, z)`, so `W_j = D` exactly (the upper bound `W_j ≤ D` is the total-degree bound: each `e_j` contains each `x_i` to power at most 1). This also confirms premise P2 of the lead-4 study for `m = 2, 3, 4`.

With `δ_{e_j} = j·δ_B` (the e-form bounds `C(m,j)·B^j`; the constants `C(m,j)` do not change the exponent), `Σ_j δ_{e_j} W_j = D · δ_B · m(m+1)/2 ≤ 1`:

m	polynomial	proved bound on δ_B	target	shortfall	the lead-4 study's §3.2 asymptote
2	S_3	1/6	1/2	3×	1/6
3	S_4	**1/24**	1/4	**6×**	1/24
4	S_5	**1/80**	1/5	**16×**	1/80

So the asymptote that §3.2 derived for the symmetric Jochemsz–May family is the **optimum of every lattice in `(p, f)^t` on a convex monomial set**, in the same determinant convention. The finite-level values of that table (1/55.0, 1/40.7, 1/35.5 for S_4) all lie below 1/24, as they must.

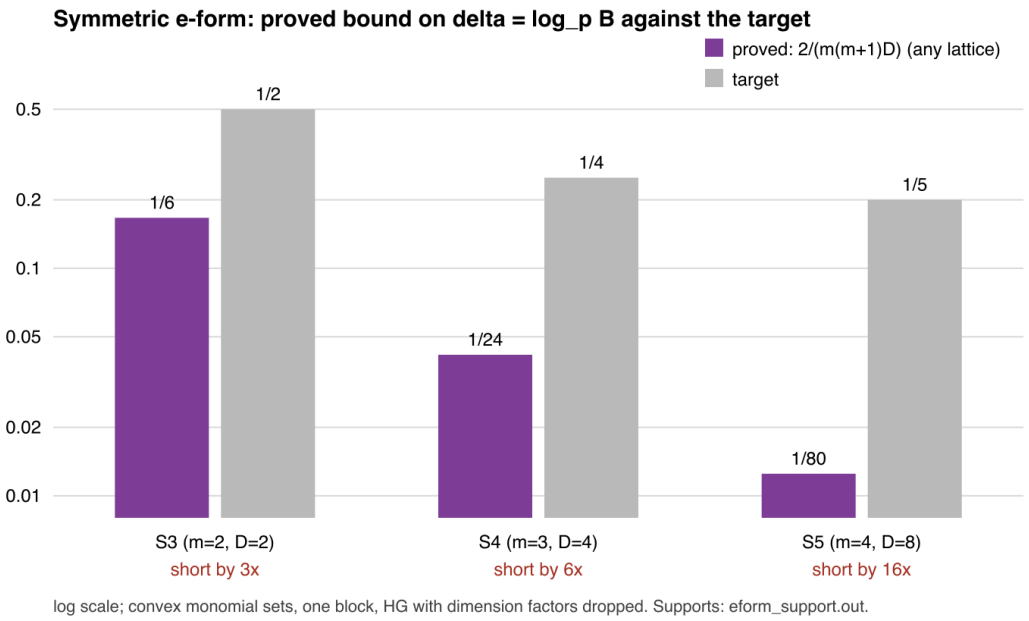


Figure 3: Proved e-form bound against the targets (log scale). Source: eform_vs_targets.svg.

The x-form, for completeness. `[measured for S₃, S₄; derived from the known degree for S₅]` In the original variables S_3 and S_4 have `W_i = 2` and 4 for every `x_i` (`xform_extents.out`); S_5 has per-variable degree 8 (Semaev). So any direction obeys $\varepsilon = \Sigma\delta_i \le 1/4$ for S_4 and $1/8$ for S_5. For equal bounds this gives $\delta \le 1/12$ and $1/32$, weaker than the volume bound of §7c (1/21.8, 1/70.8), which stays the right figure there. Theorem K's value is that it covers every direction, which the volume argument did not.

4. Checks `[measured]`

All counts are exact integers; there are no timings and no `ecbench` costs. Reproduce with `node translate_test.mjs` and `node slabs_more.mjs` (outputs `translate_test.json`, `slabs_more.json`), then `node plot.mjs`.

1. **The implementation reproduces the cited study's exact lattice gains.** `E_tr` on its 11 box cases (linear, univariate degree 2, S_3 in two and three unknowns) equals its exact `E` in **11 of 11**. Theorem K holds in each (largest $W \cdot E/c$ is 0.857, the univariate case).
2. **Property test of Theorem K.** 316 random down-closed polytopes for S_3 (120 in three unknowns, 196 in two; 2–6 facets plus axis facets, random skew): **0 violations**, largest `max_i W_i·E_tr/c_i` = 0.772. Figure 1 draws the 221 with $E > 0$ and $|M| \ge 16$.
3. **Sharpness.** Thin slabs $[0,a]\times[0,L]$ for S_3 in two unknowns (`Newt` is the box $[0,2]^2$, so `E_tr = Σ_s (a−2s+1)(L−2s+1)` in closed form, checked against the brute-force count): $W \cdot E/c_1$ rises to **0.991** at `a = 128, L = 32768`, i.e. to the bound 1 in the limit $L \gg a \gg 1$.
4. **Supports** (§3), and **Lean** (§6).

5. The three open cases of §7c

case	status	why
Unequal bounds (the symmetric e-form)	**closed** for convex `M`, one block, determinant (HG) convention	Theorem K and its corollary; the e-form supports are measured
Lattices using several targets' polynomials at once	**nothing can be said here**	The proof needs the dimension of `f̃^s·F_p[x] ∩ span M` to be at most a translate count, which comes from Ostrowski for a **principal** ideal. For `(f₁, f₂)^s` an element is a sum `Σ h_J f^J` in which supports can cancel, and `Newt` of a sum is only contained in the hull of the union. The count has no such bound, and the layer step (regular sequence) also changes. I make no claim either way.
Non-convex monomial sets	**nothing can be said here**	The fibre step needs `y + s·ν_min ∈ M`. Ostrowski only gives `y + s·ν_min ∈ conv(M) ∩ Z^n`. Running the proof with `M̄ = conv(M) ∩ Z^n` bounds `E(M) ≤ c_i(M̄)/W_i`, but the Howgrave–Graham weight is `c_i(M) ≤ c_i(M̄)`, so the ratio `E(M)/c_i(M)` is not controlled when `M` is much sparser than its hull.

6. Lean proof map

`lean/EFormReach.lean`, core Lean 4.34.1 (`lean/lean-toolchain`), no Mathlib. Build and axiom audit: `lean/lean-build.txt` (command: `lean EFormReach.lean` with `#print axioms` appended). All declarations are in the namespace `EFormReach`. No `sorry`, no custom axiom, no `native_decide`; the declarations depend on `propext`, `Classical.choice`, `Quot.sound` at most.

step	statement	Lean declaration	status
layer bound and Ostrowski (§0)	`E(M) ≤ E_tr(M)`	none	**not formalized** (inherited; needs polynomial rings over `F_p`, Newton polytopes, regular sequences)
1. fibre step	`#{y ∈ A_s : y_i=t} ≤ min(ℓ(t+sm), ℓ(t+sM'))`	none	**not formalized** (needs `Finset` of lattice points; core Lean has none)
2–3. pairs and layer-cake	`E_tr ≤ Σ_λ P_w(S_λ)`	none	**not formalized**
4. counting lemma	`w · #{pairs a positive multiple of w apart} ≤ Σ S`	`class_count_le`, `pairs_le_sum`	**checked** (strictly decreasing lists of naturals, any `w > 0`)
corollary	from `W_i·E ≤ c_i` and `Σ n_i c_i ≤ D·E`, `Σ n_i W_i ≤ D`	`reach_aux`, `reach_le`	**checked** (naturals, scaled `δ_i = n_i/D`)
e-form instances	`6u ≤ Dn`, `24u ≤ Dn`, `80u ≤ Dn` for `δ_B = u/Dn`	`eform_S3`, `eform_S4`, `eform_S5`	**checked**, given `W_j·E ≤ c_j` as hypotheses
targets out of reach	`¬ (Dn ≤ 4u)`, `¬ (Dn ≤ 5u)`	`s4_`, `s5_target_unreachable`	**checked**
e-form supports	`W_j = D`	none	computed in GP (§3), not a Lean statement

This is a **partial formalization**: Theorem K itself stays a written proof, and the reported results are conditional on it and on the inherited layer bound.

7. Limits

- **Convention.** Howgrave–Graham with dimension factors dropped, as in the cited studies. This favours the attacker, so it is a valid upper bound on reach in the determinant model. It says nothing about a solver that is not a lattice of polynomials in `(p, f)^t`.
- **One block, convex `M`.** §5 lists what is not covered. The joint three-block `(4,2)` lattice of the cited study's §5 is a multi-ideal case.

- **Sharpness is for the translate count.** Figure 1 shows the count, not a constructed lattice.
- **e-form constants.** `C(m,j)` in the bounds is ignored, as in the cited study.
- **No measured cost.** No `ecbench` run, no §5a band, no benchmark entry, no dashboard point.

8. Next checks

- Update `LEAD-ICP-coppersmith-smallroot-semaev`: its `revisit_when` names "an unequal-bound allocation beats 1/2 per block", which Theorem K answers for convex sets; the open triggers are the joint (multi-ideal) lattice and non-convex sets. (Atlas edit not made here; it is a separate PR in `crypto-autoresearcher`.)
- A direct attack on the multi-ideal case: bound `dim (f₁,f₂)^s ∩ span M` for two generic polynomials.
- Formalize steps 1–3 with a `Finset`-based library (needs Mathlib or `crypto-lean`).

9. Files

- `translate_test.mjs`, `slabs_more.mjs`, `plot.mjs`: tooling (exact counts and figures).
- `eform_support.gp`, `xform_extents.gp` and their `.out`: supports.
- `reach_sharpness`, `eform_vs_targets`, `proof_diagram` (`.svg` and `.png`): figures.
- `lean/`: Lean file, toolchain pin, build log.
- `report.pdf`: this README.

Provenance: written and run by a Claude Sonnet 5.5 research fork in the 2026-10-29 session; Lean 4.34.1, PARI/GP 2.17.3, Node v26.7.0. The proof, the checks and the Lean file are this note's own work.